Assembly Monitoring System
Complete Beginner-Friendly Code Explanation

1. Purpose

This document explains the full assembly monitoring code line by line in simple language for beginners.

2. System Overview

The system uses a camera and AI (YOLO) to monitor assembly steps and validate correct sequence execution.

3. Libraries

Libraries such as OpenCV, PyAV, Flask, threading, and YOLO provide vision processing, streaming, concurrency, and AI detection.

4. AI Models

A pose model detects human keypoints while a detection model identifies assembly components and tools.

5. Global Variables

Variables track assembly progress, detection history, sequence errors, recording state, and hand validation.

6. Multithreading

Video capture runs in a background thread while AI processing and streaming run in parallel.

7. Flask Server

Flask exposes endpoints for live viewing, resetting progress, and toggling recording.

8. frameInf Logic

This function processes each frame, runs AI detection, validates steps, and draws visual overlays.

9. Sequence Validation

Steps must be detected multiple times and in the correct order to be marked complete.

10. Hand & Spring Detection

Hand keypoints inside predefined regions confirm spring and plunger installation.

11. RTSP Capture

PyAV decodes the RTSP camera feed using GPU acceleration and crops the assembly region.

12. Live Streaming

Frames are encoded as JPEG and streamed using MJPEG for browser compatibility.

13. Program Flow

The system initializes models, starts threads, processes frames, and streams results live.

14. Conclusion

This project combines AI with rule-based logic to reliably validate industrial assembly processes.